

Dear NSSSIP Selection Committee,

I was thrilled when I found out about the opportunity to intern for the University of Maryland and the Army Research Laboratory. As a sophomore computer engineering major at the University of Maryland, College Park, I believe that I can leverage my relevant interests and skills to provide great value to the research projects in NSSSIP.

My interests lie in robotics, machine learning, and radio frequency (RF) engineering. I'm interested in how sensing, signal processing, and machine learning can be combined to enable autonomy in drones and robots, as well as how machine learning can be used to find trends in data that allow for scientific advancements.

My interests are what led me to apply to the Autonomy for Unmanned Systems, Electromagnetic Field Sensing, and Tunable RF Devices projects.

I currently work as an Embedded Software Radar Engineer for Ideal Innovations Incorporated, supporting the Army Research Laboratory, where I'm implementing a direction of arrival estimation system on the NVIDIA RUDI edge computing system using a software defined radio. I'm using Python on Ubuntu Linux to collect RF data and implement the MUSIC algorithm for RF signal direction estimation.

Last summer, I interned at the Army Research Laboratory as an RF intern where I used MATLAB to program a portable network analyzer, to transmit and receive signals at specific frequencies and power levels. I conducted lab and field tests, including mounting my setup onto a ground vehicle and scanning targets. There was a significant amount of data analysis required in terms of windowing, filtering out unwanted noise, and Fourier analysis to create an image of the target scanned. This experience not only strengthened my MATLAB skills, but also exposed me to the research process and importance of collecting "clean data" so the following data analysis process provides better results.

In 2024, I interned at the Johns Hopkins Applied Physics Laboratory where I was able to delve into a computer vision project focused on detecting lunar craters. I trained the YOLOv8 computer vision model to detect lunar craters using training data that I synthetically generated in MATLAB. Ultimately, I was able to significantly improve crater detection accuracy of the computer vision model by testing the model, identifying its shortcomings and generating additional training data to rectify the shortcomings. This experience gave me substantial exposure to deep learning as a concept and also taught me how deep learning algorithms "learn" new information and "see" patterns.

I have also gained C programming experience through the Introduction to Computer Systems class I took in my previous semester at the University of Maryland where I was able to create my own Unix-style shell and implementation of a custom dynamic memory allocator (malloc). Through this class, I've also gained a deeper understanding and appreciation of low-level programming concepts such as memory management, process control, and hardware-software interaction.

Collectively, these experiences have given me a strong foundation in programming (MATLAB, Python, C), data analysis, Linux, machine learning, and RF signal processing. I look forward to using these skills while continuing to expand my skillset.

Through this internship, I hope to deepen my understanding of my project's research area and get a better understanding of whether I'd like to pursue that field for a career. I also

seek to further hone my programming and problem solving skills which will be applicable to any field I decide to pursue. In addition, I want to form good professional connections with my internship mentors and fellow interns that I can reference later for advice and support.

Thank you for considering my application. I look forward to the opportunity to discuss the projects at NSSIP, and my qualifications in more detail.

Sincerely,
Pratheek Ramakrishna
pramakri@terpmail.umd.edu

Pratheek Ramakrishna

Clarksburg, MD | (301)693-9522 | pratheek.ramakrin@gmail.com | U.S Citizen

EDUCATION

University of Maryland, College Park

Expected: May 2028

Bachelor of Science, Computer Engineering

Cumulative GPA: 3.521

Relevant Courses: Intro to Computer Systems, Discrete Signal Analysis, Intro to Differential Equations and Linear Algebra, Chemistry for Engineers, Intro to Engineering Design

WORK EXPERIENCE

Ideal Innovations Incorporated | Adelphi, MD

Embedded Software Radar Engineer

Sept 2025 – Present

- Implementing a Python-based direction of arrival estimation system on NVIDIA RUDI using USRP X310 software defined radio and the MUSIC algorithm for real-time radio frequency direction finding

Adelphi Laboratory Center – DEVCOM Army Research Laboratory | Adelphi, MD

RF Summer Intern

Jun 2025 – Aug 2025

- Developed a MATLAB-based radar system on a FieldFox network analyzer, applying signal processing and RF analysis techniques, and integrated it onto a ground vehicle for real-world testing
- Supported international partner organizations in evaluating the performance of novel radio frequency hardware and signal processing techniques in a cost-effective manner

Johns Hopkins Applied Physics Laboratory | Laurel, MD

Academic Year Intern

Oct 2023 – May 2024

- Developed 500 detailed lunar surface images in MATLAB, incorporating 2700+ annotations of lunar craters for use in model training and analysis
- Trained YOLOv8 machine learning model using Python to detect lunar craters using synthetic images

Summer Intern

July 2023 – Aug 2023

- Compared camera calibration software for NASA's Dragonfly prototype using MATLAB and Python, resulting in enhanced prototype performance
- Presented findings to Dragonfly's navigation team, informing navigation system design

PROJECTS

Over-Terrain Vehicle

Jan 2025 – May 2025

- Collaborated with a team of eight to design and build an autonomous vehicle capable of navigating an obstacle course and extinguishing candles
- Coded navigation algorithm in C++ on Arduino and completed the robot's electrical wiring

TECHNICAL SKILLS

Programming Languages: MATLAB, Verilog, Java, Python, C, C++

Tools: Oscilloscope, Function Generator, Spectrum Analyzer, Network Analyzer, Logic Analyzer, Soldering

Software: Unix, Git, KiCAD, Simulink, Xilinx Vivado, Cadence PSpice

ACTIVITIES

Terrapin Rocket Team - Avionics Team Member

Sept 2025 – Present

- Developed a custom PCB flight computer with an onboard IMU and pressure sensor using KiCAD to capture real-time telemetry for a Level 1 high-power rocket
- Implemented C++ firmware on Teensy 4.1 microcontroller to interface with sensors